Problem 1. Let $a \in \mathbb{R}$. Prove that if $0 < a < 1$, then $0 < a^2 < a$. (Notice that $0 < a < 1$ means that both $0 < a$ and $a < 1$.)

You can *only* use the axioms on the "Axioms" handout and results that we have proved that appear on that document.

Proof. Let $a \in \mathbb{R}$ such that $0 < a < 1$. This means that $0 < a$ and $a < 1$. We will show first that $0 < a^2$ and then that $a^2 < a$, and this will show that $0 < a^2 < a$.

Since $0 < a$, by definition, we know that $a - 0 \in \mathbb{R}^+$. By the additive inverse axiom, $a - 0 = a$, so $a \in \mathbb{R}^+$. By positive closure, $a^2 = a \cdot a \in \mathbb{R}^+$. Therefore $a^2 = a^2 - 0 \in \mathbb{R}^+$, which means precisely that $0 < a^2$.

Since $a < 1$, by definition, $1 - a \in \mathbb{R}^+$. Since we already know that $a \in \mathbb{R}^+$, therefore $a(1-a) \in \mathbb{R}^+$ by positive closure. Notice that

$$\begin{aligned} a(1-a) &= a(1) + a(-a) && \text{by distribution} \\ &= a + a(-1)a && \text{by multiplicative identity and HW 3 \#2} \\ &= a - a^2 && \text{by commutativity and HW 3 \#2} \end{aligned}$$

Therefore $a - a^2 \in \mathbb{R}^+$, which means that $a^2 < a$. This completes the proof. □

Problem 2. Let $x \in \mathbb{R}$. The **absolute value** of x is defined by

$$|x| = \begin{cases} x & \text{if } x \geqslant 0 \\ -x & \text{if } x < 0 \end{cases}.$$

Show for all $x, y \in \mathbb{R}$ that $|xy| = |x| \cdot |y|$.

Proof. Let $x, y \in \mathbb{R}$. We will show that $|xy| = |x| \cdot |y|$.

First observe that if $x = 0$, then $xy = 0y = 0$ and $|x| = |0| = 0$. Therefore $|xy| = |0| = 0 = 0|y| = |x| \cdot |y|$. If $y = 0$, the result holds as well. We will consider the following other cases.

Case 1: Both x and y are positive. Thus $|x| = x$ and $|y| = y$. By positive closure, xy is also positive. Thus $|xy| = xy = |x| \cdot |y|$.

Case 2: One of x and y is positive and one is negative. Without loss of generality, assume that x is positive and y is negative. Thus $|x| = x$ and $|y| = -y$. Similarly, xy is negative, so $|xy| = -xy$. Therefore $|xy| = -xy = x \cdot -y = |x| \cdot |y|$.

Case 3: Both x and y are negative. Thus $|x| = -x$ and $|y| = -y$. Furthermore, xy is positive. Thus $|xy| = xy = -x \cdot -y = |x| \cdot |y|$.

This completes the proof. □

Problem 3. In this problem, you will show that $\sqrt{2} + \sqrt{3}$ is irrational.

1. Prove that if $x \in \mathbb{R}$ is irrational and $m \in \mathbb{Q}\backslash\{0\}$, then xm is irrational.

   Proof. Let $x \in \mathbb{R}\backslash\mathbb{Q}$ and $m \in \mathbb{Q}\backslash\{0\}$. We will argue by contradiction, so we assume that $mx = \frac{p}{q}$ for some $p, q \in \mathbb{Z}$ with $q \neq 0$. Since $m \neq 0$, it has a multiplicative inverse m^{-1}. Therefore $m^{-1}mx = m^{-1}\frac{p}{q}$. Therefore $x = m^{-1}\frac{p}{q}$, and by closure of the rationals, this implies that x is rational. This is a contradiction; therefore mx must be irrational. □

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2. Prove that $\sqrt{6}$ is irrational.

   Proof. We follow the same roadmap as the proof that $\sqrt{2}$ is irrational. We write $\sqrt{6}$ as a rational number p/q with p the least possible natural element, which is possible by the Well-Ordering Principle. Then $p^2 = 2 \cdot 3 \cdot q^2$. So we can deduce that p^2 is even and, by a theorem

proved in class, also p is even, say $p = 2k$ for some positive integer k, which is smaller than q. Now, from $p^2 = 2 \cdot 3 \cdot q^2$ we obtain $2k^2 = 3q^2$, which implies that $3q^2$ is even. Now we want to deduce that also q^2 is even, so we use congruence modulo 2. Since $1 \equiv 3 \pmod 2$, we have $q^2 \equiv 3q^2 \equiv 0 \pmod 2$, thus q^2 is even. Again, this means that q is even, say $q = 2\ell$. Since $p = 2k$ and $q = 2\ell$, we get $\sqrt{6} = \frac{p}{q} = \frac{k}{\ell}$, which contradicts that p is the smallest possible denominator. Thus $\sqrt{6}$ is irrational. □

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3. Prove that $\sqrt{2} + \sqrt{3}$ is irrational.

   Proof. We will prove this statement by contradiction. We assume that $\sqrt{2}+\sqrt{3} \in \mathbb{Q}$. By closure of the rationals, this says that $(\sqrt{2} + \sqrt{3})^2 \in \mathbb{Q}$. Notice that

   $$(\sqrt{2} + \sqrt{3})^2 = 2 + 2\sqrt{6} + 3 = 5 + 2\sqrt{6}.$$

   Rearranging the above equation, we see that $2\sqrt{6} = (\sqrt{2} + \sqrt{3})^2 - 5$, so $2\sqrt{6} \in \mathbb{Q}$ by closure of the rationals. But by parts (a) and (b) of this problem, we know that $2\sqrt{6}$ is irrational, so this is a contradiction. Therefore $\sqrt{2} + \sqrt{3}$ is irrational. □

Problem 4. For each of the following proposed theorems, provide a proof if the theorem is true, or provide a counterexample if it is false.

1. **Theorem (?)**

   Let $a, b \in \mathbb{Z}$. If $ab \equiv 0 \pmod 3$, then either $a \equiv 0 \pmod 3$ or $b \equiv 0 \pmod 3$.

   Proof. We will prove this theorem is true by contrapositive. We assume that $a, b \in \mathbb{Z}$ and $a \not\equiv 0 \pmod 3$ and $b \not\equiv 0 \pmod 3$. The Division Algorithm says that a and b are each congruent to either 1 or 2 mod 3. We will consider the following cases.

   Case 1: One of a and b is congruent to 1 mod 3. Without loss of generality, we will assume that $a \equiv 1 \pmod 3$. Therefore $ab \equiv 1b \pmod 3$ by a theorem from class (which says if $x, y, z, w \in \mathbb{Z}$, $n \in \mathbb{N}$, $x \equiv y \pmod n$, and $z \equiv w \pmod n$, then $xz \equiv yw \pmod n$). Thus $ab \equiv b \pmod 3$, and since $b \not\equiv 0 \pmod 3$, this means that $ab \not\equiv 0 \pmod 3$ by the Division Algorithm.

   Case 2: Both a and b are congruent to 2 mod 3. Using the theorem referenced in Case 1, we see that $ab \equiv 2 \cdot 2 \pmod 3$, i.e. $ab \equiv 1 \pmod 3$. Therefore $ab \not\equiv 0 \pmod 3$ by the Division Algorithm.

   In either case, we've deduced that $ab \not\equiv 0 \pmod 3$. This completes our proof by contrapositive.

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2. **Theorem (?)**

   Let $a, b \in \mathbb{Z}$. If $ab \equiv 0 \pmod 4$, then either $a \equiv 0 \pmod 4$ or $b \equiv 0 \pmod 4$.

   Solution. This statement is false. If $a = b = 2$, then $ab \equiv 0 \pmod 4$. However, the Division Algorithm shows us that both a and b are congruent to 2 mod 4 and hence not congruent to 0 mod 4. □

Problem 5. For which natural numbers is $n! > 3^n$? Prove it.

Theorem. For all natural numbers $n \geqslant 7$, $n! > 3^n$.

Proof. We proceed by induction.

Base case. When $n = 7$, $n! = 5040$ and $3^n = 2187$ so in this case, $n! > 3^n$.
Inductive step. Suppose for some $k \geqslant 7$ that $k! > 3^k$. Then, since $k + 1 > 0$, we have

$$(k+1)! = k!(k+1) > 3^k(k+1).$$

And since $k + 1 > 3$, we have

$$3^k(k+1) > 3^k \cdot 3 = 3^{k+1}.$$

By transitivity, we conclude $(k+1)! > 3^{k+1}$ so by induction, we obtain the desired result. □

Problem 6. Is the following proposition true or false? Prove or give a counterexample.
For each nonnegative integer n, $8^n \mid (4n)!$.

Solution. This is true.

Proof. We proceed by induction.
Base case. For the base case, when $n = 0$, $8^0 = 1$ and $(4n)! = 1$ and $1 \mid 1$ since $1 = 1 \cdot 1$.
Inductive step. Now suppose that for some $k \geqslant 1$, we know that $8^k \mid (4k)!$. Then there exists an integer j such that $(4k)! = 8^k \cdot j$. Therefore,

$$\begin{aligned}(4(k+1))! = (4k+4)! &= (4k+4)(4k+3)(4k+2)(4k+1)4k! \\ &= 8^k \cdot j \cdot (4k+4)(4k+3)(4k+2)(4k+1) \\ &= 8^k \cdot j \cdot 4 \cdot (k+1)(4k+3) \cdot 2 \cdot (2k+1)(4k+1) \\ &= 8^{k+1} \cdot j \cdot (k+1)(4k+3)(2k+1)(4k+1)\end{aligned}$$

and by the closure of the integers, $j \cdot (k+1)(4k+3)(2k+1)(4k+1) \in \mathbb{Z}$. Hence, $8^{k+1} \mid (4(k+1))!$ and by induction we obtain the desired result. □

Problem 7. 1. For $n \in \{1, 2, \ldots, 7\}$, fill out the following table.

n	$1^3 + 2^3 + \cdots + n^3$	$n(n+1)$
1	1	2
	etc.	

2. Conjecture an equality relating $1^3 + 2^3 + \cdots + n^3$ and $n(n+1)$.

   Solution. By looking at the first seven rows of the table, we conjecture that

$$1^3 + 2^3 + \cdots + n^3 = \left(\frac{n(n+1)}{2}\right)^2.$$

3. Prove that your conjecture is correct.

   Proof. We will prove the conjecture by induction on n.

   Basis Step. Let $n = 1$. Then $n^3 = 1^3 = 1 = \left(\frac{1(1+1)}{2}\right)^2 = \left(\frac{n(n+1)}{2}\right)^2$. Thus the statement holds for the base case.

   Induction Step. Suppose that for some $k \in \mathbb{N}$, that the formula holds for k, that is, assume

$$1^3 + 2^3 + \cdots + k^3 = \left(\frac{k(k+1)}{2}\right)^2.$$

We will show that it holds for $k+1$. Observe that

$$\begin{aligned}
1^3 + 2^3 + \cdots + k^3 + (k+1)^3 &= \left(\frac{k(k+1)}{2}\right)^2 + (k+1)^3 \quad \text{by our induction hypothesis} \\
&= \frac{k^2(k+1)^2}{4} + \frac{4(k+1)^3}{4} \\
&= \frac{k^2(k+1)^2 + 4(k+1)^3}{4} \\
&= \frac{(k+1)^2[k^2+4k+4]}{4} \\
&= \frac{(k+1)^2(k+2)^2}{4} \\
&= \left(\frac{(k+1)((k+1)+1)}{2}\right)^2.
\end{aligned}$$

Therefore the statement holds for $k+1$ too. This completes our induction step.

Thus by induction, the theorem holds for all $n \in \mathbb{N}$. □

Problem 8. Prove that the sum of the cubes of any three consecutive natural numbers is divisible by 9. (For example, $4^3 + 5^3 + 6^3 = 405 = 9 \cdot 45$.)

Proof. We will prove this statement by induction. Specifically, we will show that 9 divides $n^3 + (n+1)^3 + (n+2)^3$ for all $n \in \mathbb{N}$.

Basis Step. Let $n = 1$. Then $n^3 + (n+1)^3 + (n+2)^3 = 1^3 + 2^3 + 3^3 = 1 + 8 + 27 = 36 = 4 \cdot 9$, so the statement holds for $n = 1$.

Induction Step. Suppose that for some $k \in \mathbb{N}$ that 9 divides $k^3 + (k+1)^3 + (k+2)^3$, i.e., $k^3 + (k+1)^3 + (k+2)^3 = 9m$ for some $m \in \mathbb{Z}$. We will show that 9 divides $(k+1)^3 + (k+2)^3 + (k+3)^3$. Notice that

$$\begin{aligned}
(k+1)^3 + (k+2)^3 + (k+3)^3 &= (k+1)^3 + (k+2)^3 + (k^3 + 9k^2 + 27k + 27) \\
&= k^3 + (k+1)^3 + (k+2)^3 + 9k^2 + 27k + 27 \\
&= 9m + 9k^2 + 27k + 27 \quad \textbf{by induction hypothesis} \\
&= 9(m + k^2 + 3k + 3).
\end{aligned}$$

Since $(m + k^2 + 3k + 3) \in \mathbb{Z}$, this shows that 9 divides $(k+1)^3 + (k+2)^3 + (k+3)^3$.

This completes our proof by induction. □

Problem 9. Complete Exercise 18 (a) and (b) on page 184 of the textbook (*Mathematical Reasoning* by Sundstrom).

Solution.

(a) This proof has some *issues*. First note that the statement they call $P(n)$ is not the whole formula. It's unclear if they've even shown that the base case is true thanks to this. The inductive step starts out okay, but then the **cardinal sin** is that they appear to assume that the statement holds for $P(k+1)$! This is very bad proof writing, and we must not fall into temptation. I will critique the next proof before writing a correct and well-written proof.

(b) Also not great. The $P(n)$ notation is incorrect, and it's lead to a bigger mess than the $P(n)$ issue above. Statements are true or false (they certainly don't evaluate to an integer—this isn't a comp sci course), so the basis step is all wrong. The inductive step has the same issue, which is that $P(n)$ is just defined incorrectly. Notice that the arithmetic is presented in a way better way.

Notice for both of these: I'm not saying that there are errors in the arithmetic here—the bigger issue is with the proof writing. We are trying to write *clear* and *logically* sound proofs, and both of the provided proofs do not meet these criteria. Here's a better proof.

Proposition. For each natural number n, $1 + 4 + 7 + \cdots + (3n - 2) = \frac{n(3n-1)}{2}$.

Proof. We will prove this theorem by induction.

Basis Step. Let $n = 1$. Then $1 = \frac{1(3\cdot 1-1)}{2}$, so the statement holds for $n = 1$.

Inductive Step. Let $k \in \mathbb{N}$ and assume that the statement holds for k, i.e., we assume that

$$1 + 4 + 7 + \cdots + (3k - 2) = \frac{k(3k - 1)}{2}.$$

We will show that the statement holds for $k + 1$. Observe the following, with the first equality given by our induction hypothesis:

$$\begin{aligned} 1 + 4 + 7 + \cdots + (3k - 2) + (3(k + 1) - 2) &= \frac{k(3k - 1)}{2} + (3(k + 1) - 2) \\ &= \frac{k(3k - 1)}{2} + \frac{2(3(k + 1) - 2)}{2} \\ &= \frac{k(3k - 1) + 2(3(k + 1) - 2)}{2} \\ &= \frac{3k^2 + 5k + 2}{2} \\ &= \frac{(k + 1)[3(k + 1) - 1]}{2} \end{aligned}$$

Therefore the statement holds for $k + 1$ as well. This completes the inductive step.

By induction, we've shown that $1+4+7+\cdots+(3n-2) = \frac{n(3n-1)}{2}$ for all $n \in \mathbb{N}$. This completes the proof. □

Problem 10. Do not turn these in, but I recommend doing the progress checks for sections 3.4 and 4.1 in your textbook.